
8th International Conference on Computer Science and Computational Intelligence (ICCSCI 2023)

Philippine Court Case Summarizer using Latent Semantic Analysis

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Abstract

Artificial Intelligence in the field of Law has introduced a lot of innovations to aid legal experts, and one of those are summarizing systems. Text summarization is the process of condensing text into a compact summary highlighting the important parts of the original text. As years pass by, numerous text summarization systems are being developed, and improvements keep getting made. These advancements have been helping humans in easing labor work with summarizing lengthy documents. With this, the researchers would like to propose a tool that will summarize legal documents, specifically cases obtained from the Philippine Supreme Court website. This includes extracting meanings and analyzation of words, and similarity of sentences from the original legal document obtained from the website of the Philippine Supreme Court. The tool will take the content of the original document, analyze the words, structure, and sentences according to context, and finally produce the summary. The proposed tool will be called SUMMIT, from the words “Summarize It”. The method to be used to summarize the documents using the tool is called Latent Semantic Analysis (LSA). The researchers will use Python as their programming language in developing the summarizer tool.

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Peer-review under responsibility of the scientific committee of the 8th International Conference on Computer Science and Computational Intelligence 2023

Keywords: Natural Language Processing; Latent Semantic Analysis; Extractive-based summarization

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1. INTRODUCTION

Artificial Intelligence in Law involves the application of computer and mathematical techniques to make law and legal texts more understandable, manageable, accessible, or even predictable [1]. The application of AI concepts in Law began roughly during the 1970s to 1990s; many of the projects relating to these concepts are focused on formally modeling legal arguments in a computer-processable form, and computationally modeling legislation and legal rules. As the years passed by, since the 2000s, the application of AI in Law has evolved from knowledge-representation techniques toward machine learning-based approaches. One of the major applications was focused on the litigation discovery or the process of obtaining evidence for a lawsuit. Traditionally, litigation discovery was performed manually by lawyers who would read each document and indicate if the document is relevant or irrelevant to the case. Because of the advent of Artificial Intelligence, a computer-based document-review technique was developed. This technique aimed to automatically distinguish the relevant and irrelevant documents in the litigation discovery.

According to Susskind [2], on his work “The End of Lawyers: Rethinking the Nature of Legal Services”, AI really takes off in the 2020s. AI and other technologies are enabling machines to take on many of the tasks that a lot of people used to think required human lawyers, and this seems to be happening frequently. He even added that AI applications in the legal scene will eventually warm up by 2020. Aligned to this, as ever larger amounts of legal documents had been produced and been available electronically, the legal industry has led to a need for systems that can automatically summarize this increasing number of documents.

Text summarization is the process of condensing text into a compact summary highlighting the important parts of the original text. Radev, Hovy, and McKeown defined a summary as a “text that is produced from one or more texts, that conveys important information in the original texts, and that is no longer than half of the original texts and usually significant less than that” [3]. As years pass by, numerous text summarization systems are being developed, and improvements keep getting made. These advancements have been helping humans in easing labor work with summarizing lengthy documents.

Focusing on the field of law, legal experts are responsible for clerical work that requires accuracy and speed. These lawyers are responsible for understanding, interpreting, and researching wide variety of legal documents. Since case summaries and digests are usually created manually, it often takes a lot of work time and human expertise. Developing an automatic text summarizer specifically for court case files would help in reducing these costs while providing accurate summaries. This type of system will also be beneficial not only in the field of law, but also in other industries as the generation of data rises rapidly. That is also the reason why this specific field in NLP and AI has been popular for researchers nowadays.

In the Philippine setting, despite of the continuous AI advancements, there is still a small number of research in this field considering the rapidly increasing number of legal documents, especially court case documents, each year. In support to that, the Supreme Court of the Philippines, which is the highest court in the Philippines and has the exclusive power to promulgate rules concerning the protection and enforcement of constitutional rights, and legal assistance to the underprivileged, has released its Judicial Annual Report last 2019. [4] The statistical findings show that out of 6, 014 judicial cases, 5, 792 cases are disposed by the end of the year which represents 96% of production rate.

In this study, an automated method for summarizing legal texts will be created using legal cases from the Philippine Supreme Court. The researchers considered concentrating on summarizing these types of court documents that may aid students and attorneys to save time in summarizing these court records that they may utilize in their study and as a basis for developing arguments to future relevant cases.

Moreover, while legal narration style can provide valuable clarity and precision in conveying legal information, it may need to be adapted to suit the unique requirements of building an AI. Balancing formal language with accessibility, context understanding, and ethical considerations becomes vital to create AI systems that effectively interact with users and align with broader societal needs.

This study reflects the following literatures:

Due Diligence Incorporating AI

Typically, the traditional due diligence process, especially in legal investigation, is not only time-consuming but also labor-intensive, which can take a long time to be completed. An application of AI to the process can provide a swifter and decreased labor time in reviewing documentation. Incorporating AI in due diligence facilitates a more accurate identification process with less human interaction and a cheaper option for clients. The process of machine-generated summarization can also be mirrored with this kind of system in terms of lesser human labor, with ensured accuracy and quality of machine performance [5].

Text Summarization in PYTHON: Extractive vs. Abstractive Techniques Revisited

Inclined with the rapid development of technology and bloom of the internet comes the inevitable generation of vast data and information. Be it structured and unstructured, or in the form of images, videos, or simple, plain texts. Text summarization, the task of evaluating large pieces of texts and generating intelligent and accurate summaries, poses a significant relevance in the field of NLP and research [6].

In the legal field, summarization of legal case judgements and texts has been a major problem because of the length and complexity of each document. Many summarization algorithms have been proposed up to this date including both for general text documents and some specifically focused on the legal text documents among various countries.

Text Summarization within the Latent Semantic Analysis Framework: Comparative Study

In this paper, Badry, Eldin, and Elzanfally [7] presented a comparative study among different algorithms based on the Latent Semantic Analysis (LSA) approach. The researchers presented five (5) algorithms with the LSA approach and compared all of them with each other to show which algorithm has more gaps, is more usable, and more suitable given the type of document to be summarized. The researchers aimed to identify the best algorithm that can be used in general forms of documents and does not have too many disadvantages or lacking.

All of the presented algorithms take only single-documents for inputs. All algorithms had the same main steps except of Cross and Topic which have the pre-processing step. These algorithms are based on each other, and the significant difference is the process taken to fill in the gap of the base algorithm. Badry et al. hoped for graph-based approaches to be developed in the future which can still be used along with LSA to enhance the performance of summarization systems.

Understanding Automatic Text Summarization-I: Extractive methods

Extraction-based summarization in simple terms involves taking the most important lines from the input document and combines it to create the output summary [3]. Since it does mainly rely on deep learning, all lines and words that are placed in the summary actually belongs to the original document. Roy stated that extractive summaries tend to work better than its abstractive counterpart because extractive methods does not require semantic representation and natural language generation.

Extraction-based summarizers work by creating an intermediate representation first, wherein its task is to determine the most important information from the texts that will be summarized based on the said representation. The most common representation used by extractive summarizers is the topic representation. Topic representation can be further classified into multiple approaches, depending on its implementation: topic word, frequency-driven approach, LSA, and others.

LetSum, An Automatic Legal Text Summarizing System

Farzindar and Lapalme [8] decided to create an automated text summarization system to process this increasing number of legal documents. The researchers developed LetSum for summarizing the legal record of the proceedings of federal courts in Canada; they have presented the summary in a table-style format. The researchers chose to create this law-specific automated text summarizer because of the complexity of the domain. There are some specific terminologies in the legal domain that may produce ambiguities. With that, they have constructed their own conceptual dictionary containing 200 concepts of legal domain. The researchers based the summarization on identifying the thematic structure to find the important and argumentative themes of the judgement.

Extractive vs. NLG-based Abstractive Summarization of Evaluative Text: The Effect of Corpus Controversiality

There are two general approaches when it comes to text summarization: Abstraction and Extraction. Carenini and Cheung [9] stated that abstraction involves generating summaries through learning the corpus and observing contexts between sentences. On the other hand, extraction is the method that extracts words and phrases from the

original texts and concatenates the extracts to form a summary.

Latent Semantic Analysis

Latent Semantic Analysis (LSA) is a technique for creating vector-based matrices regarding the semantic representation— words, sentences, or phrases from a document and compute the similarity of the text pairs within the matrix by comparing the vectors [10]. LSA was developed mainly for information retrieval and query for large database of documents.

Hastings provided the main processes of using LSA and how each stage functions. The said processes are listed below:

1. Collection of large set of texts and separation of the texts into “documents”. Each paragraph from the whole input text is treated as an individual document (since sentences within paragraphs are assumed to be coherent and related).
 2. Creation of matrices for documents and terms. The documents identified within the former process are placed into the x values of the matrix and the terms-without repetition, within all of the documents-are correspondingly placed into the y axis. Terms are defined as words that are present in multiple documents.
 3. Scoring of cells. Each value in individual cells can be assigned significant weights to reduce the effect of common words appearing through the corpus. This process may also include the removal of stop words.
 4. Singular Value Decomposition (SVD) is applied with a parameter to specify the number of needed dimensions.
- The aforementioned processes will produce a total of three matrices. First, one having k-singular values. Second, one having a k-dimensional vector for each term. And third, one having a k-dimensional vector for each document.

Using Latent Semantic Analysis in Text Summarization and Summary Evaluation

Steinberger and Jezek’s technique [11] called ‘lengthy method’ has the same first two steps as the previously mentioned approach, but in this case, the lengthy method avoided the limitations of Gong and Liu’s approach. Murray et al. also tried to avoid Gong and Liu’s disadvantages. Their approach involved two main functions which are the ability to select more than one sentence from the topmost concept and to determine how many sentences will be accumulated from each concept using the Σ matrix.

Generic Text Summarization Using Relevance Measure and Latent Semantic Analysis

Gong and Liu conducted a study that produced two text summarization methods with different approach. The first one uses an Information Retrieval method that ranks the sentences by relevance [12]. The second method on the other hand uses latent semantic analysis to determine semantically important sentences. This study was created in order to test and compare the two proposed methods and determine whether these methods will be an efficient summarizer that can be used to a document with a with range whilst having low percentage of redundancy.

Single-Document Summarization Using Latent Semantic Analysis

On a separate study by Dokun and Celebi [13] both the systems of Gong and Liu [12] and Steinberger [11] were studied. The researchers adopted the style from the said existing methods and proposed an improved version of Gong and Liu’s approach. The avesvd and ravesvd methods were extensions to Gong and Liu’s, and it uses a sentence selection method based on Steinberger and Jezek’s study. It uses the similar processes starting from the decomposition and the creation of matrix for the SVD. Then comes the proposed modifications for sentence selection. The avesvd method selects one sentence from the average of most important sentences among all concepts. Then the second sentence is chosen from the second most important sentences and continues until all predefined number of sentences are collected. The ravesvd method on the other hand has similar functions but focuses more on reducing the dimension of the concept while not losing any important sentences after the reduction process.

Text summarization using Latent Semantic Analysis

Ozsoy and Alpaslan explained and assessed the different LSA-based summarization algorithms and proposed two new algorithms namely Cross and Topic for summarizing both Turkish and English documents. This study also presented a comparison between LSA-based algorithms and machine learning-based algorithms to identify their differences [14].

Based on the implementation of these LSA-based summarization algorithms, it has been discovered that for most of the datasets used in this study, the Cross method has been the best approach. This approach only failed to summarize the third dataset of Turkish texts for it includes short documents that hinder the method to perform well. On the other hand, the Topic method has also performed well for it had produced the same result as the Murray, Renals and Carletta’s Method. With regard to the comparison of LSA-based algorithm to the machine learning-based

algorithms, it is said that LSA-based algorithms do not perform well compared to the machine learning-based algorithms that use outer information.

It is proven in this study that both Cross and Topic methods can be used effectively in summarizing English texts. It is also proven that, unlike the machine learning-based algorithms, LSA-based algorithms only use information from the input document. This claim was supported by the findings of the previous studies which state that LSA is unsupervised and do not need external information to extract semantic information that exists in a document.

NLP-based Latent Semantic Analysis for Legal Text Summarization

On a separate study by Merchant and Pande [15], LSA was also used for legal text summarization. The natural language processing approach was used in the proposed automatic text summarization to generate summaries from lengthy judgements both from single and multi-document. The structure of the text summarizer mainly composed of three parts, namely pre-processing, the LSA model, and sentence selection. In the preprocessing phase, only the judgement of the whole case will be taken from the input document. It will include the names of the people involved, cases cited, headnote, and the verdict. In this step, abbreviations were considered as stop words and are removed in order to avoid errors in sentence selection. After the LSA model finalizes its analysis, the sentence selection process is executed in order to determine the sentences that will be included in the summary.

2. METHODOLOGY

2.1 Dataset

The data used and gathered are the court case documents from the official website of Supreme Court of the Philippines. The said data will be used as the input for the summarizer. The researcher used three types of data. The first dataset are those court case documents that are from the official website of Supreme Court of the Philippines, thus, the aforementioned data will be converted to its text file format using the software application of Batas.org. The reference summaries created by legal professionals make up the second dataset, which was used to assess the tool's summaries' accuracy. Ultimately, the third data are the evaluation of the legal professionals to the questionnaire provide by the researchers.

2.2 System Architecture

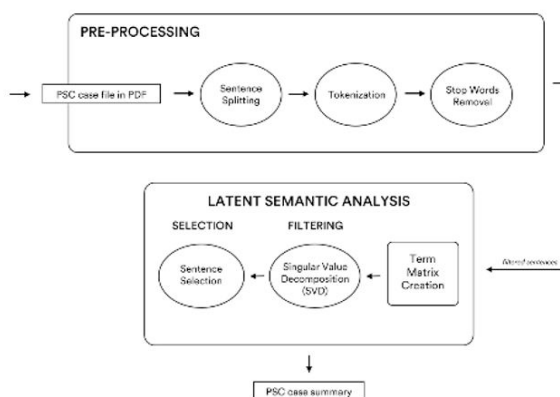


Fig 1. System Architecture

The entire flow of the system is depicted in Fig 1.

1. Preprocessing

In the pre-processing stage, the input document will be broken down into sentences followed by

terms/words/token extraction. This stage includes several processes such as sentence splitter, where paragraphs are divided into sentences; tokenization, where the sentences are split into words, tokens, phrases, and other significant elements that must not be omitted; and stop words removal, where the stop words such as ‘a’, ‘an’, ‘the’, etc. are removed to increase the accuracy of the analysis.

2. Filtering

The system will use the Latent Semantic Analysis (LSA) technique as the NLP approach to weigh the relevant information for each part. In LSA, the main concept is that it extracts the input document and converts it to a term-sentence matrix and filters it through the use of Singular Value Decomposition (SVD). SVD will help the tool to find the relationship between words/tokens and sentences and disregard that irrelevant information.

3. Selection

After the filtering phase, the tool will now produce a list of best candidate units or sentences that could be included in the summary. In this phase, the tool will rank the filtered sentences through their scores. The sentence-term that has the highest scores will be included in the summary.

2.3 Data Treatment

The LSA technique will be applied to the data or input text to implement the summarization process. The output or summary acquired from the process will generate a compression ratio in comparison to the original document. This compression rate will be the basis of the result to see if the system has actually shortened the input text, and produced a relevant summary.

$$\text{Compression Ratio} = \frac{\text{number of words/sentences in the summary}}{\text{number of words/sentences in the source document}}$$

3, RESULTS AND DISCUSSIONS

After the development process, the researchers implemented the summarization system on fifteen (15) court case documents. Then, the data needed for determining the compression ratio for both the word and sentence count were gathered. The following illustrations will demonstrate how the system shortened the length of the input document while maintaining relevant information present in the generated summary. The following results were acquired:

1. Compression Rating in terms of Words

Court Case Documents	Number of Words in the Summary	Number of Words in the Source Document	Word Compression Ratio
1. G.R. No. 198529	682	5272	0.12346
2. G.R. No. 210920	1461	5441	0.26852
3. G.R. No. 224595	591	2855	0.20701
4. G.R. No. 225741	436	3410	0.12786
5. G.R. No. 227704	1975	7829	0.25227
6. G.R. No. 229084	795	4290	0.18531
7. G.R. No. 230221	739	3771	0.19569
8. G.R. No. 232380	1248	3909	0.31926
9. G.R. No. 232393	1500	5238	0.28636
10. G.R. No. 241247	2097	5913	0.35464
11. G.R. No. 241834	500	4539	0.11016
12. G.R. No. 242165	671	3692	0.18174
13. G.R. No. 243926	878	3512	0.25000
14. G.R. No. 248130	1280	4673	0.27391
15. G.R. No. 248264	685	3118	0.21969
Average Word Compression Ratio			0.25572 or 25.57%

Fig 2. Compression Rating for Words Results

Fig 2 shows the general register number for each court case document used for testing the system. The table also presents the number of words in both the summary and the source document, the word compression ratio of each respective document, and the average word compression ratio

2. Compression Rating in terms of Sentences

Court Case Documents	Number of Sentences in the Summary	Number of Sentences in the Source Document	Sentence Compression Ratio
1. G.R. No. 198529	59	247	0.23887
2. G.R. No. 210920	67	268	0.25000
3. G.R. No. 224595	37	151	0.24503
4. G.R. No. 225741	41	164	0.25000
5. G.R. No. 227704	75	303	0.24752
6. G.R. No. 229084	43	175	0.24571
7. G.R. No. 230221	44	182	0.24176
8. G.R. No. 232380	43	179	0.24022
9. G.R. No. 232393	80	327	0.24465
10. G.R. No. 241247	76	306	0.24837
11. G.R. No. 241834	40	169	0.23669
12. G.R. No. 242165	49	196	0.25000
13. G.R. No. 243926	42	170	0.24706
14. G.R. No. 248130	55	221	0.24887
15. G.R. No. 248264	35	147	0.23810
Average Sentence Compression Rate			0.24486 or 24.49%

Fig 3. Compression Rating for Sentences Results

Fig 3 similarly shows the general register number for each court case document. It also presents the number of sentences in both the summary and the source document, the sentence compression ratio of each respective document, and the average word compression ratio.

The proponents of this study were able to acquire the data for the word and sentence ratio by gathering the word and sentence count from the system's output, then manually computing the said values to determine compression ratio for both of the categories. The tables shown above shows the individual word and sentence compression ratio for each document. After tallying the results for the compression ratio, the compression rate in terms of words got an average of 25.57%. The sentences on the other hand tallied an average of 24.29% compression rate.

4. RESULTS AND DISCUSSIONS

The compression ratio measures how shorter the summary when compared to the original document. As a convention, low compression ratio is considered to be the ideal values, in contrast to high compression ratio. Since the average compression rate for the words and sentences are 25.27% and 24.29% respectively, the tool managed to reduce the input documents size by more or a less a quarter. The researchers therefore conclude that the summary generated by the tool has a decent quality in terms of length. However, since the court case documents have different formats and partitions, the coherence and context of the summary's information have some issues. The researchers found out that some sentences in the summary have little to no connection to their preceding and succeeding sentences. This may decrease the summaries quality in terms of coherence. This issue may also be due to the fact that the LSA approach has a difficulty in summarizing long documents [7].

In line with the results, the proponents of this study recommend future researchers to include a part segmentation phase after the pre-processing of the input document. The part segmentation phase may dissect the document into title, facts, issues, ruling, and opinions, which are the common parts of a Philippine court case document. By allowing the documents to be summarized for each part, the results may encourage a more coherent and concise summary.

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